TEXORA · LECTURE SHEET

Purl Structure

Purl completes the four basics by alternating the loop facing course by course instead of wale by wale. It needs the most demanding needle in weft knitting — the double-headed needle transferred between beds — and repays it with lengthwise elasticity, thick reversible effects and the garter and moss textures of knitwear.

TE-3205 · FABRIC STRUCTURE AND DESIGN · TOPIC 12 · LEVEL 3 TERM 2

1. The structure

In purl (links-links), **whole courses alternate**: one course is drawn entirely to the face, the next entirely to the reverse. The notation makes the family resemblance and the difference plain at a glance — rib alternates *along* the row, purl alternates *between* rows:

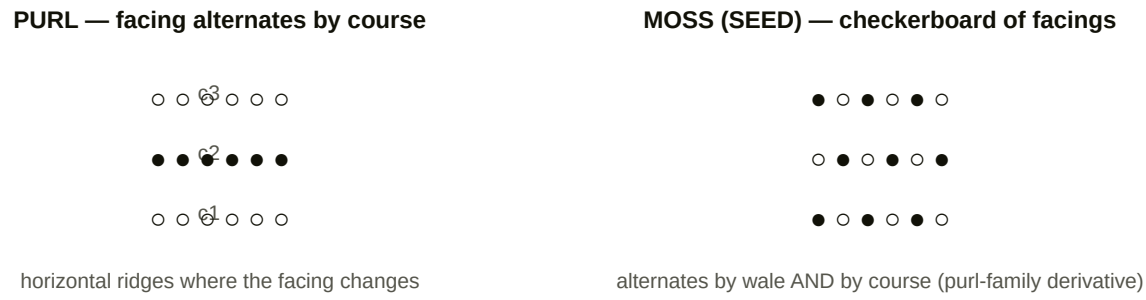


Figure 1 — Dot notation for purl (left) and its moss derivative (right), in the course convention of Topic 9. Read the purl panel by rows: the whole row changes symbol from course to course — the exact transpose of rib's rule.

2. How purl is knitted — the double-headed needle

A loop can only be drawn toward the side its needle works from, so changing the facing of a whole course means moving the work to needles acting from the other side. The purl machine (links-links machine) uses **double-headed latch needles** — a hook at each end — sliding in tricks shared by **two beds in line** (facing each other horizontally rather than at right angles as in rib). After each course, the needles are **transferred** bodily to the opposite bed (the hooks engaged by transfer elements), so the next course is drawn through from the other side. The mechanism is slow and exacting compared with jersey or rib production, which is why true purl is a knitwear structure, not a bulk apparel-fabric structure, and why its effects are so often *imitated* on flat V-bed machines by transferring selected stitches.

3. Properties

- **Balance in the course direction:** opposed courses cancel the bending tendency along the length, so purl does not curl lengthwise; the fabric shows soft **horizontal ridges** at each change of facing.
- **Lengthwise extensibility:** the stored slack is in the course direction (the reverse of rib): purl extends and recovers strongly in length.
- **Thickness and reversibility:** the ridged, double-faced texture is thick and warm for its yarn, and patterned purl work (garter panels, moss borders) reads on both faces — the classic sweater textures are purl-family: **garter stitch** is plain purl alternation itself; **moss/seed stitch** alternates facing by wale and course together (Figure 1, right).
- **Unravelling:** like jersey, purl can be unravelled from either end of the fabric.

4. Worked example — purl fabric weight

GSM OF A PURL KNITWEAR FABRIC

A wool-blend purl fabric, relaxed: WPC = 10 per cm, CPC = 12 per cm, stitch length 4.2 mm, yarn 29.5 tex.

GSM = WPC × CPC × l × tex ÷ 100 = 10 × 12 × 4.2 × 29.5 ÷ 100 = 120 × 4.2 × 29.5 ÷ 100 = 14,868 ÷ 100 ≈ **149 g/m²**.

Tightness: K = √29.5 ÷ 4.2 = 5.43 ÷ 4.2 = **1.29** — slack by the Topic 10 scale, as knitwear structures are normally knitted: the purl fabric gets its body from the ridged structure and the coarse count, not from tightness. The identity of Topic 10 applies unchanged — purl changes loop facing, not loop accounting.

5. Purl among the four basics — closing the classification

STRUCTURE	FACING PATTERN	BEDS	SIGNATURE PROPERTY
Plain (jersey)	All loops one way	1	Lightest; curls; ladders
Rib	Alternates by wale	2 (rib gating)	Widthways recovery; no curl
Interlock	Two ribs intermeshed	2 (interlock gating)	Both faces smooth; stable; heaviest
Purl	Alternates by course	2 in line, transfer	Lengthways stretch; ridged, reversible

With this table the structural half of the course closes: every weft-knitted fabric met in practice is one of these four, a derivative of one of them by tuck/miss/transfer (Topic 9), or a warp knit. TE-3206 now

turns both halves of the course — woven Topics 1–8 and knitted Topics 9–12 — into laboratory practice: analysing real samples and writing their reproduction specifications.

Glossary

Purl (links-links) — structure alternating loop facing course by course. **Double-headed needle** — latch needle hooked at both ends, transferable between two beds in line. **Transfer** — moving needles (or stitches) from one bed to the other to change the drawing side. **Garter stitch** — knitting term for plain purl alternation. **Moss (seed) stitch** — facing alternating by wale and by course.

Key points

• Purl = rib's rule transposed: alternation by course, not by wale; notation rows change symbol wholesale.
• It requires double-headed needles transferred between two beds in line — hence slow, knitwear-scale production.
• Properties: lengthways stretch and recovery, horizontal ridges, reversible textures (garter, moss); GSM by the same identity (worked: ≈149 g/m², K 1.29).

Practice

1. Write the dot notation for a purl fabric over courses 1–4 and 6 wales, taking course 1 as a face course.
2. A fabric extends and recovers mainly in the length and shows horizontal ridges; its loops all face one way within each course. Identify the structure and the needle type its machine must have.
3. Compute the GSM of a purl fabric with WPC 9, CPC 11, l = 4.6 mm, yarn 33 tex.
4. Why is moss stitch classed with the purl family although its notation also alternates within the course?

Answer key

1. c1: ● ● ● ● ● ●; c2: ○ ○ ○ ○ ○ ○; c3 as c1; c4 as c2 — whole rows alternate.
2. **Purl**. The machine must use **double-headed (links-links) needles** transferable between two beds in line, since each course is drawn from the opposite side to the last.
3. GSM = 9 × 11 × 4.6 × 33 ÷ 100 = 99 × 4.6 × 33 ÷ 100 = 15,028 ÷ 100 ≈ **150 g/m²**.
4. Because its defining operation is the purl one — loops drawn to both faces with course-wise changes of facing (in moss, the facing changes every course in every wale, and additionally alternates along the course). The mechanism (transfer/double-headed work or its flat-machine

imitation) and the resulting reversible, ridged handle are purl-family traits; rib's alternation, by contrast, never changes facing from course to course.